



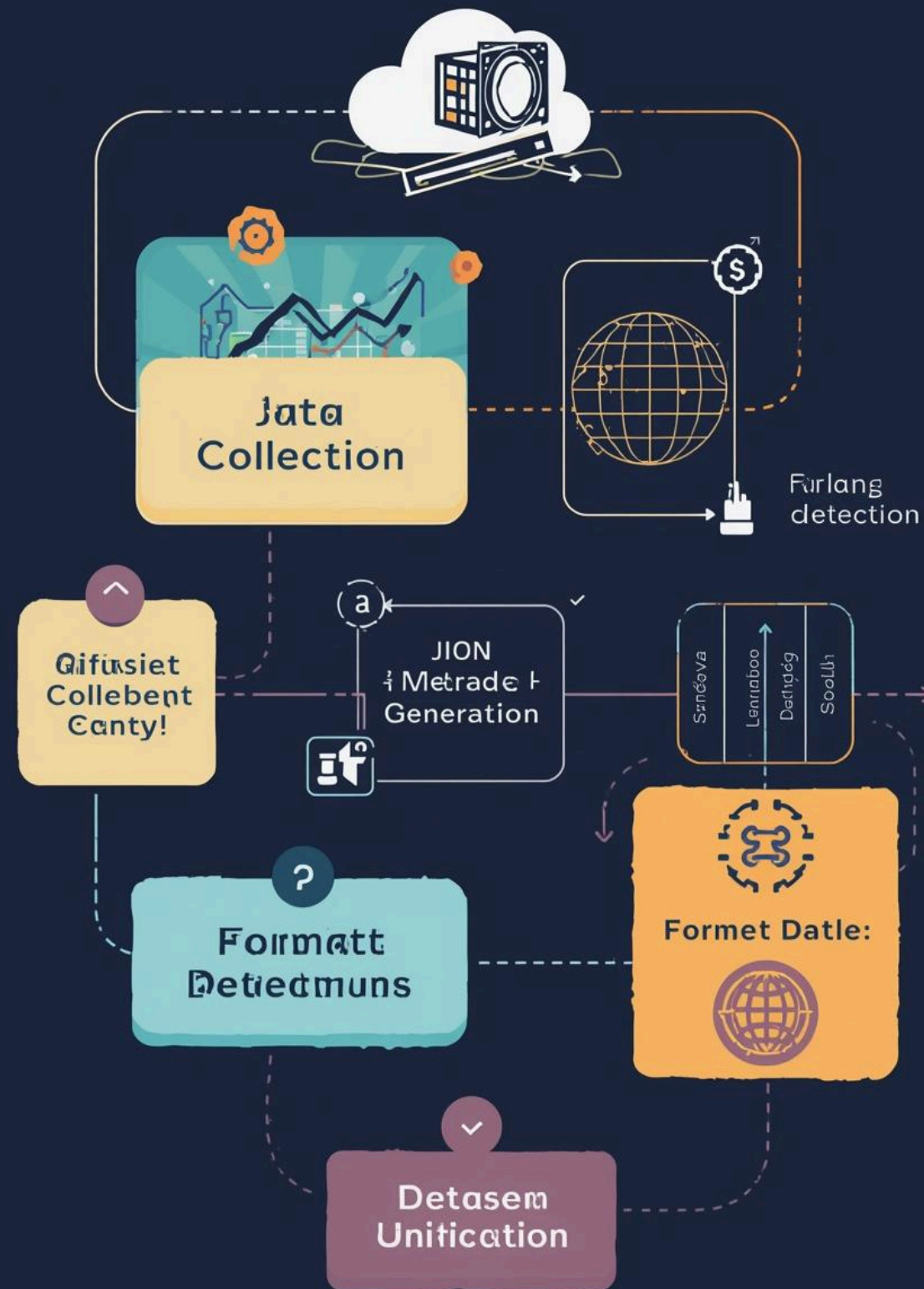
Deepfake Detection Pipeline

Integration with Blockchain Technology

Chintan Mange

Dataset Creation

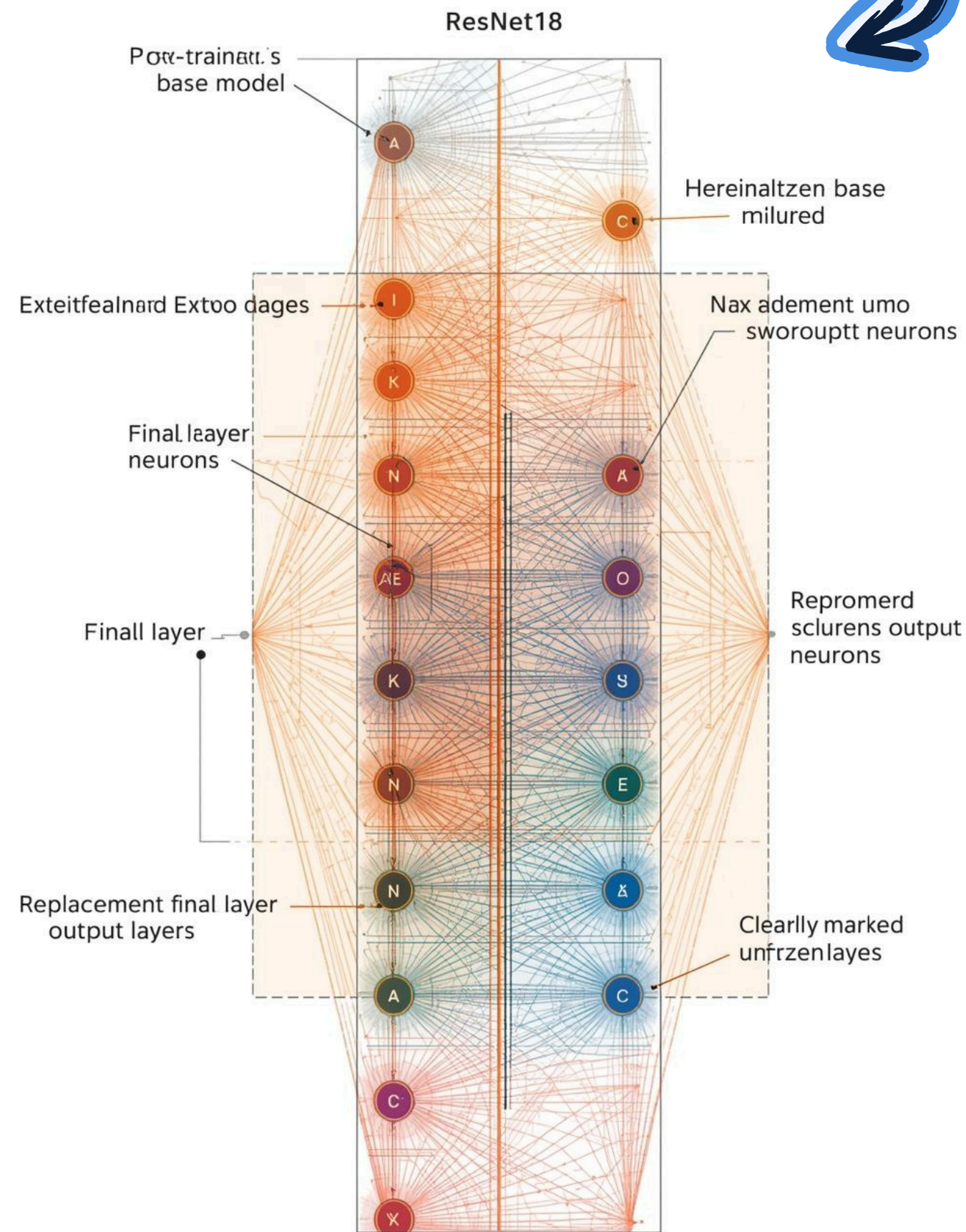
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Dataset & Preprocessing

Structuring the data for deepfake analysis

The dataset incorporates **JSON metadata** for real and fake images, enabling auto-detection of JPG/PNG formats to create a unified dataset labeled as either 0 (Real) or 1 (Fake).

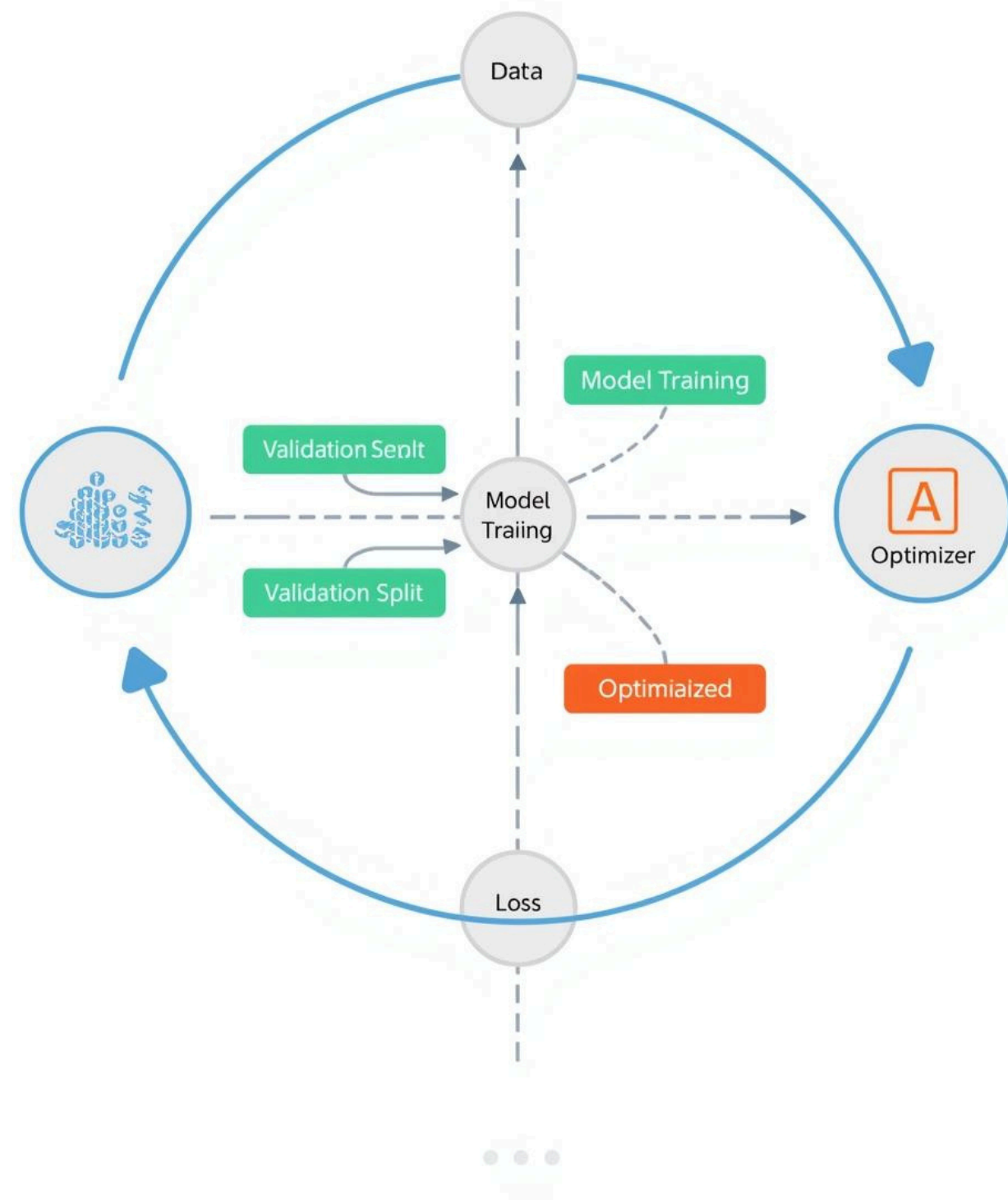


Model Architecture

Understanding ResNet18 Modifications

The model utilizes ResNet18, pre-trained on ImageNet, with modifications including a new final layer for binary classification and partial freezing of specific layers to enhance learning efficiency.

Training Loop



Training Pipeline

Understanding the training and validation process

The training pipeline consists of mild data **augmentation techniques** and an 80-20 train/validation split, utilizing CrossEntropyLoss for training and ReduceLROnPlateau for learning rate management.

Results and Accuracy Metrics



Achievements from model training process

95.75%

Validation Accuracy

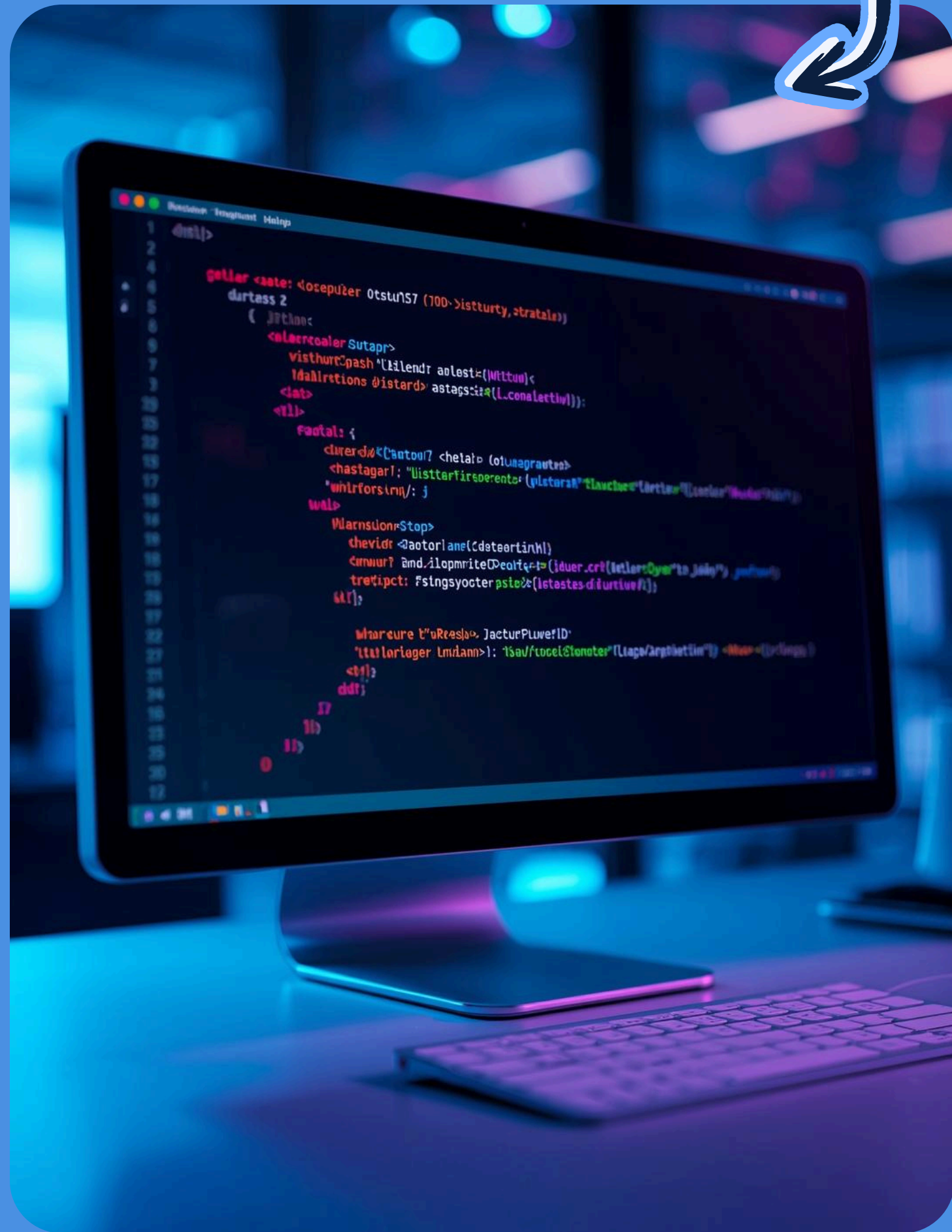
Achieved at the end of training.



12 epochs

Stabilization Epoch

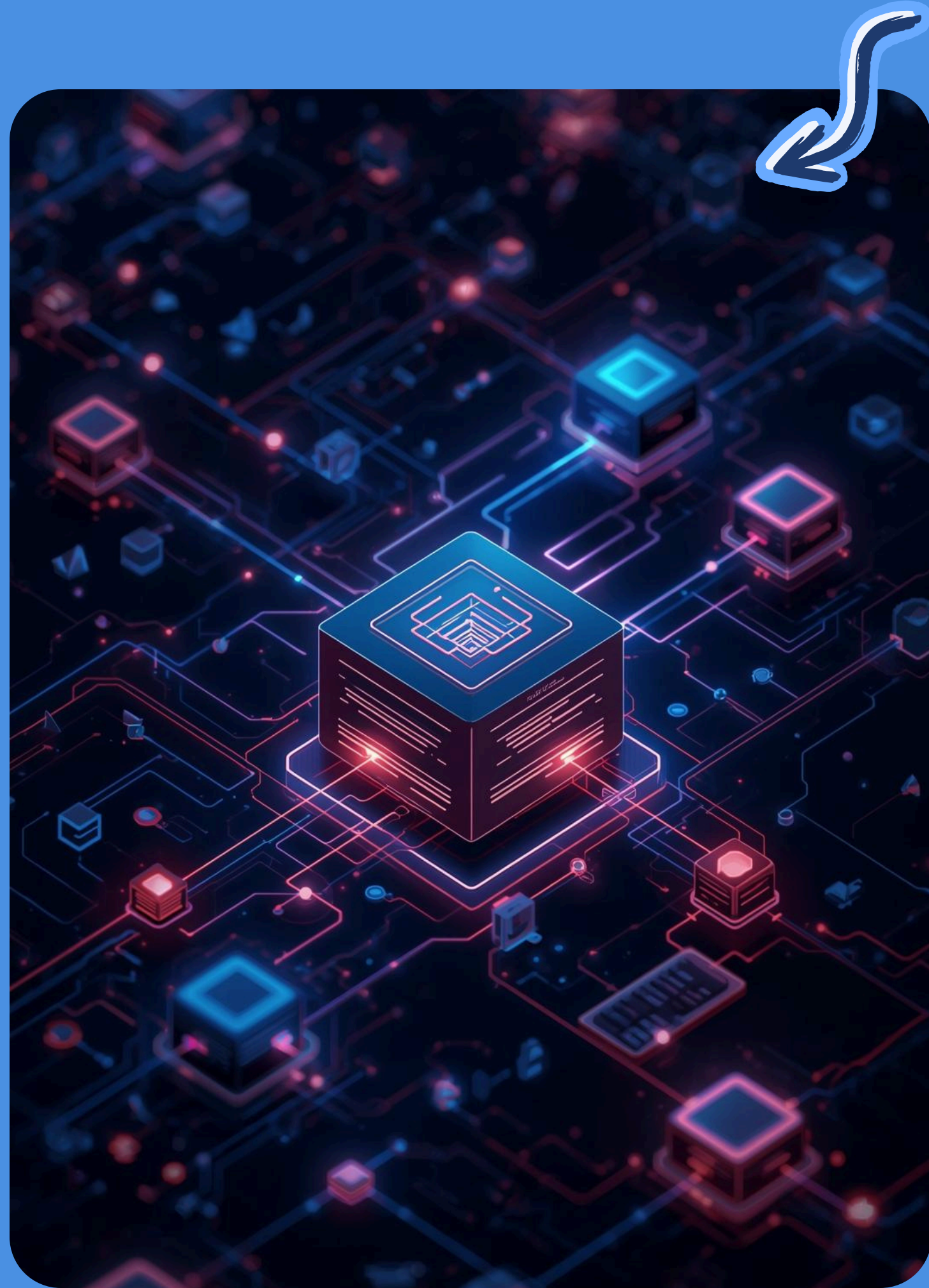
Training performance stabilized by epoch twelve.



Inference Outputs

Understanding the JSON generation process

During inference, the model predicts whether an image is real or fake, automatically generating a structured JSON output that includes the image name, prediction result, and confidence level for transparent logging.



Blockchain Integration

Securely logging deepfake predictions

Utilizing the **Ethereum** smart contract enables secure storage of predictions and hashes, ensuring data integrity and transparency through a seamless interface with Ganache and Web3.py.



Conclusion & Future Work